Report: Clustering Seismic Data Analysis

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1 Objective

The objective of this analysis is to discover meaningful seismic activity patterns using unsupervised clustering techniques on global earthquake data. Specifically, this report applies:

- K-Means clustering,
- DBSCAN (density-based clustering),
- Hierarchical (Agglomerative) clustering, and
- Gaussian Mixture Model (GMM) clustering.

The analysis aims to:

- identify natural geographic groupings of earthquakes,
- compare model behavior under different cluster assumptions,
- detect outlier/sparse seismic events,
- infer which method best captures irregular tectonic patterns.

2 Dataset Description

Primary dataset: `isc-gem-cat.csv` (ISC-GEM global earthquake catalog).
Supplementary file: `isc-gem-suppl.csv` (supporting records/metadata).

The final feature set used for clustering:

- `lat` (latitude),
- `lon` (longitude),
- `depth`,
- `smajax` (semi-major uncertainty axis),
- `sminax` (semi-minor uncertainty axis).

Usable cleaned records: 74,156 events.

3 Preprocessing Pipeline

3.1 Data cleaning

- Selected required columns based on available header names.
- Converted values to numeric with coercion of malformed entries.
- Removed incomplete rows using null filtering (`dropna()` logic).

3.2 Feature scaling

All clustering features were standardized using `StandardScaler` to zero mean and unit variance. This step is critical because raw scales differ significantly (for example, spatial coordinates versus depth and uncertainty), and unscaled features can bias distance-based algorithms.

4 Exploratory Data Analysis

4.1 Distribution and spread of seismic features

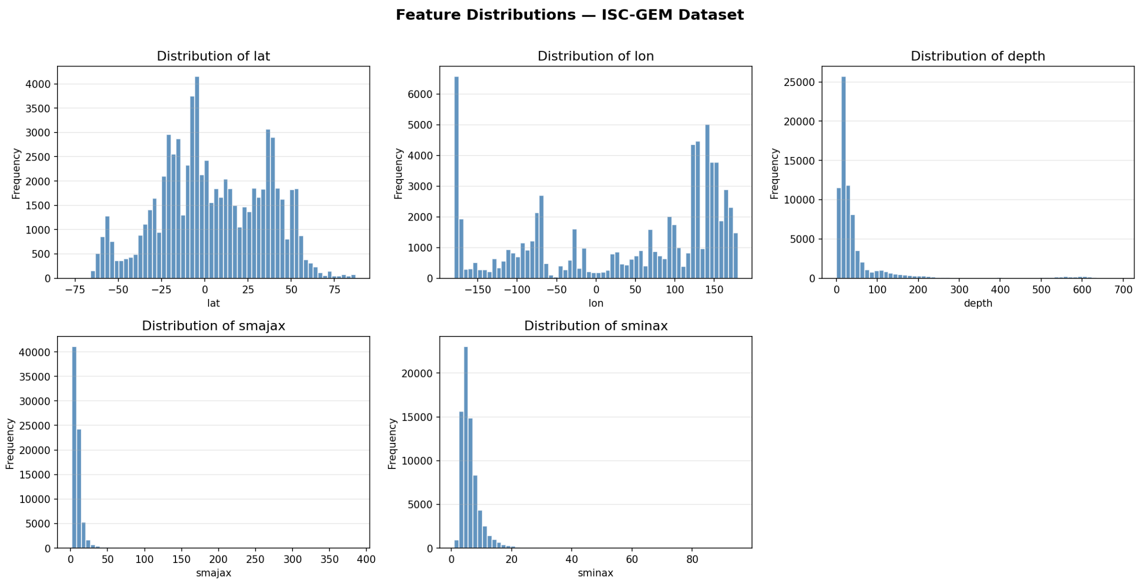

Figure 1: Distribution plots for selected seismic features.

Inference:

- Features are not perfectly Gaussian and show skewed tails.
- Depth and uncertainty features show broad spread, motivating scaling.
- Non-uniform distributions suggest simple spherical cluster assumptions may be limiting.

4.2 Feature correlation structure

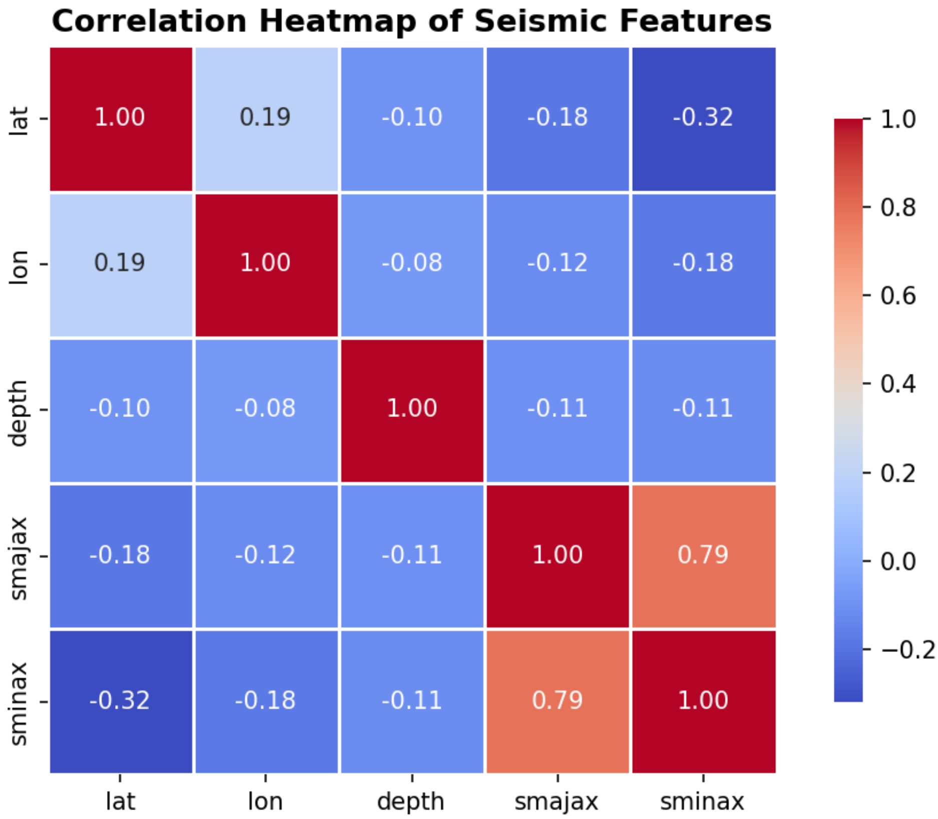

Figure 2: Correlation heatmap of seismic attributes.

Inference:

- Some features exhibit moderate dependence, while others remain weakly coupled.
- Mixed correlations justify testing multiple clustering paradigms.

5 K-Means Clustering

5.1 Elbow method selection

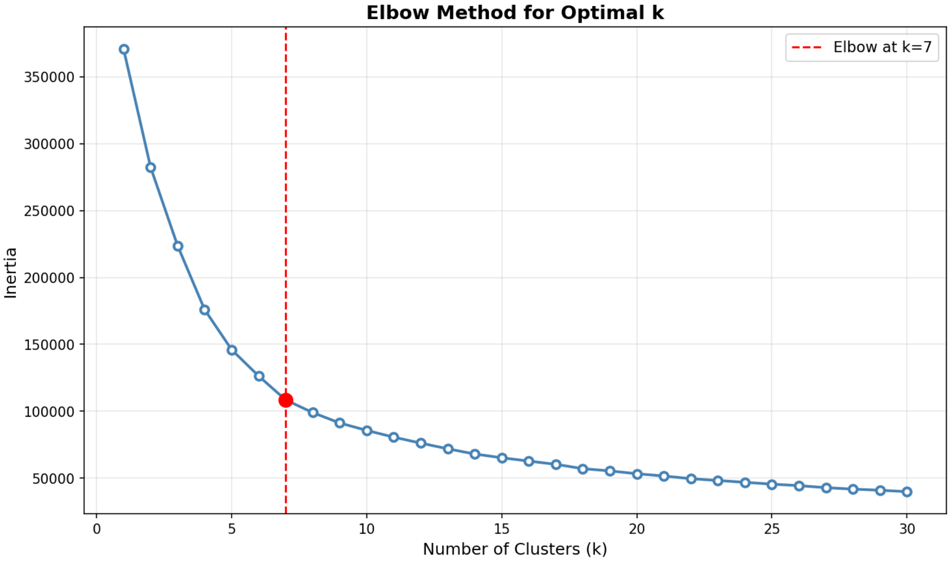

Figure 3: Elbow curve (inertia vs number of clusters) for K-Means.

Result: Optimal cluster count selected as $k = 7$.
Silhouette score (K-Means, $k = 7$): 0.3432.
Inference:

- Elbow behavior indicates diminishing returns beyond $k = 7$.
- The silhouette value indicates moderate separation, suitable for baseline segmentation.

5.2 K-Means world map

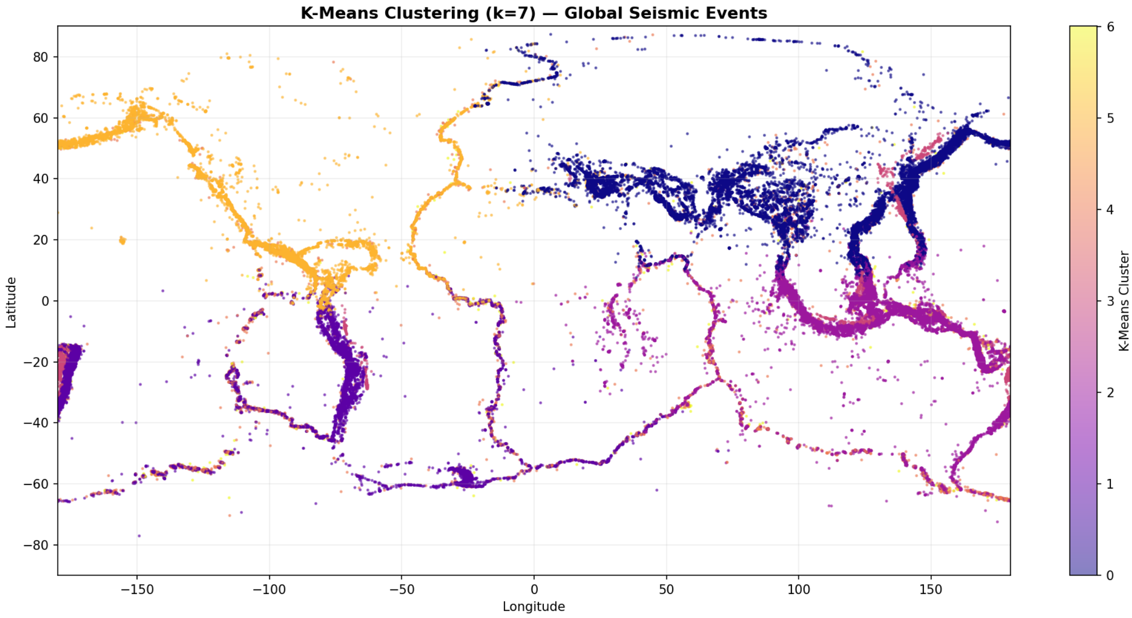

Figure 4: K-Means cluster assignments over global seismic locations.

Inference:

- Major active belts (Pacific margin, Mediterranean region, Asian collision zones) are captured.
- K-Means creates compact partitions but can merge geographically separated areas with similar feature values.
- Appropriate for broad zoning, less suitable for non-convex plate-boundary geometry.

6 DBSCAN Clustering

6.1 Nearest-neighbor distance (epsilon tuning)

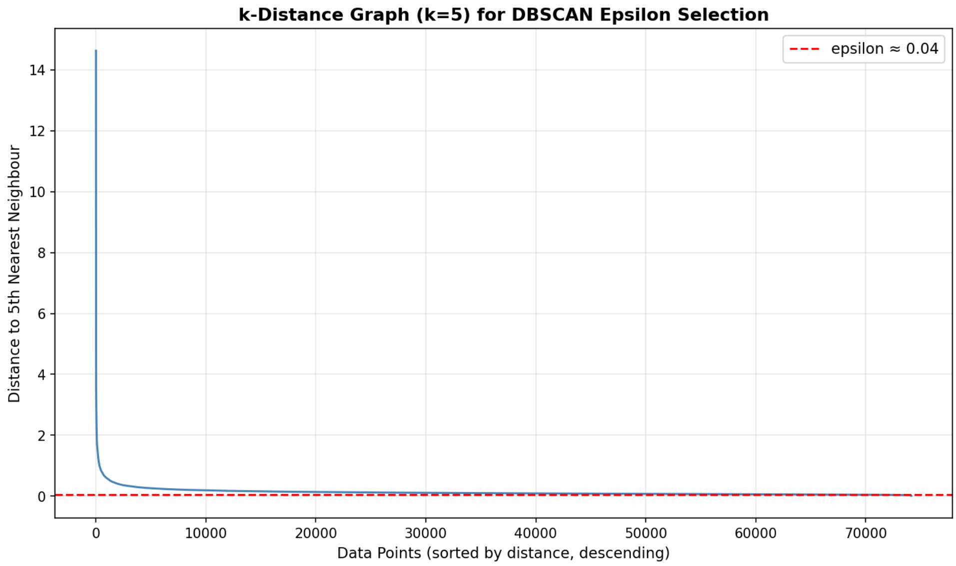

Figure 5: k-distance curve used to choose DBSCAN ϵ.

Chosen parameters: $\epsilon = 0.04$, min_samples $= 5$.

Inference:

- The knee in the distance curve marks transition from dense neighborhoods to sparse points.
- Parameter choice is sensitive: small changes in ϵ can strongly alter cluster/noise balance.

6.2 DBSCAN world map

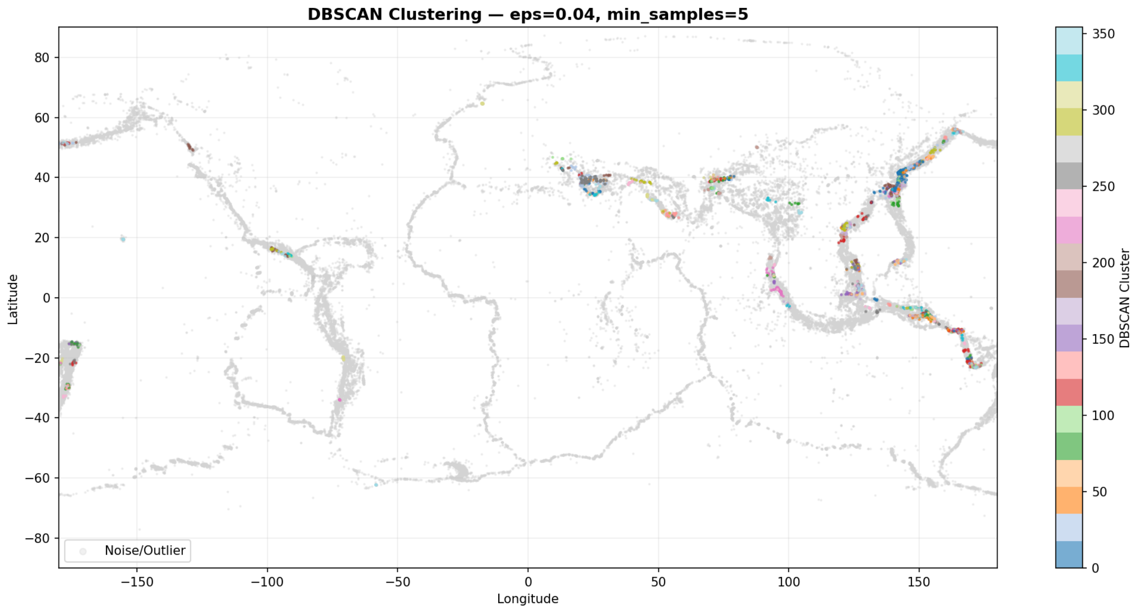

Figure 6: DBSCAN clustering map with dense regions and noise points.

Observed output statistics:

- Number of clusters found: 355
- Noise points: 70,545 ($\approx 95.1\%$)

  Inference:

- DBSCAN effectively isolates local dense seismic pockets and outliers.
- Very high noise proportion indicates current ϵ is strict for this standardized feature space.
- For a cleaner global partition, a parameter sweep (ϵ, min_samples) is recommended.

7 Hierarchical Clustering

7.1 Dendrogram representation

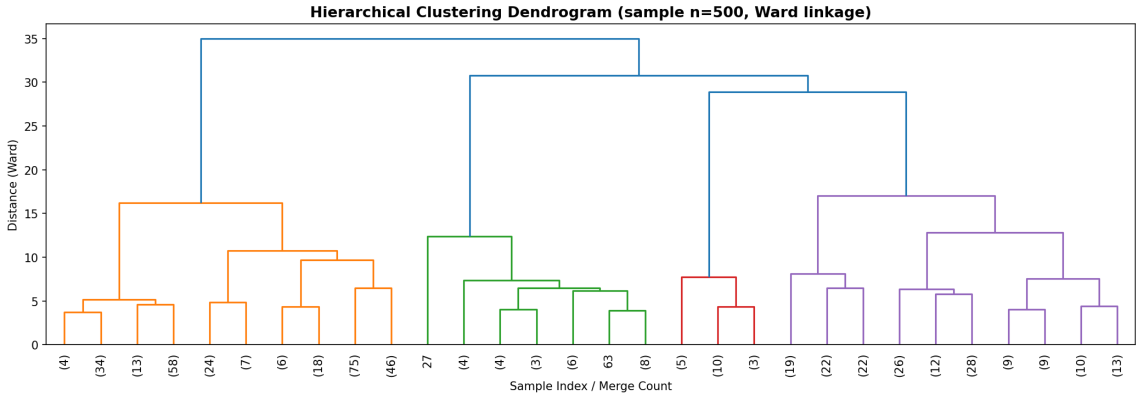

Figure 7: Hierarchical clustering dendrogram view.

Inference:

- The tree structure shows nested similarity levels among seismic observations.
- Useful for interpreting multi-scale grouping behavior.

7.2 Silhouette-based cluster selection

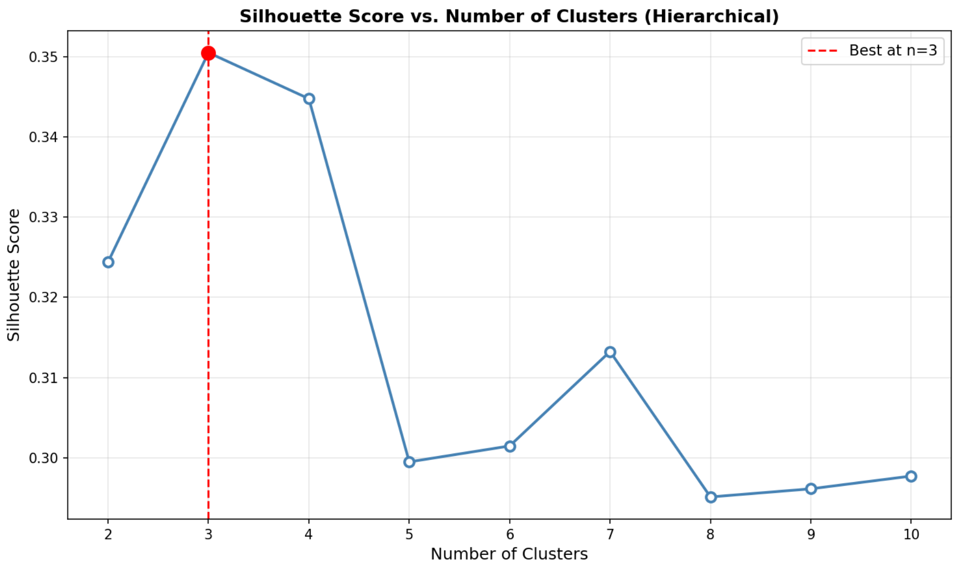

Figure 8: Silhouette score vs number of clusters for hierarchical clustering.

Observed silhouette values (sampled):

- $n = 2 : 0.3244$
- $n = 3 : 0.3505$ **(best)**

- $n = 4 : 0.3448$
- $n = 5 : 0.2995$

Estimated optimal (from silhouette): $n = 3$.
Final plotted hierarchical map in notebook: $n = 5$ (Ward linkage).
Inference:

- $n = 3$ is statistically strongest by silhouette.
- $n = 5$ may still be chosen for richer geographic interpretability (more granular regions).
- This trade-off (metric optimum vs domain readability) should be explicitly stated in evaluation.

7.3 Hierarchical world map

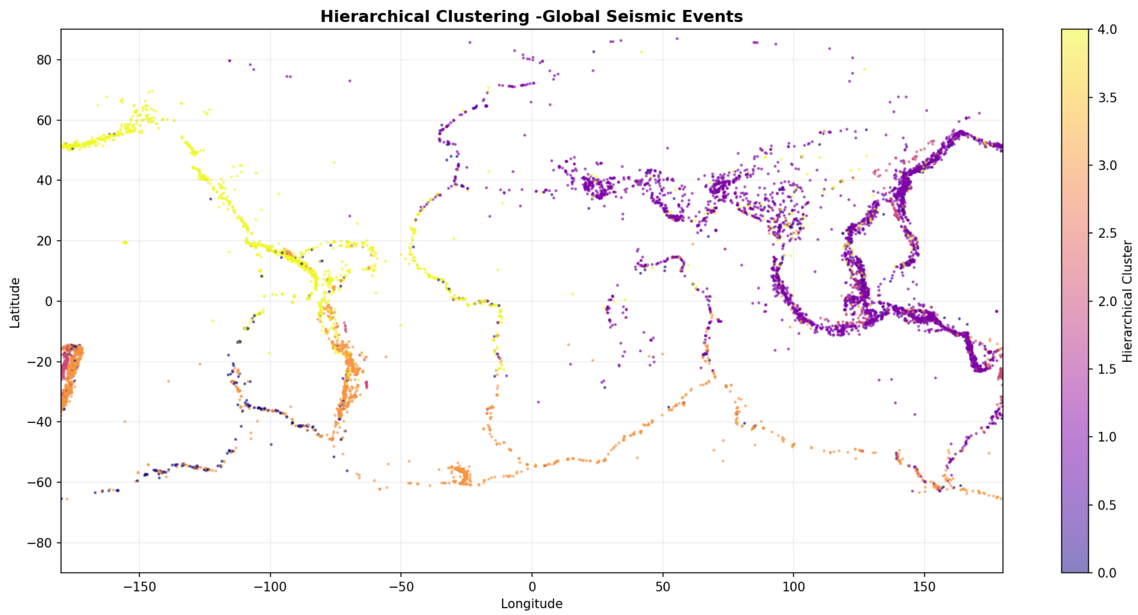

Figure 9: Hierarchical clustering world map (Ward linkage).

8 Gaussian Mixture Model (GMM)

8.1 Model order selection via information criteria

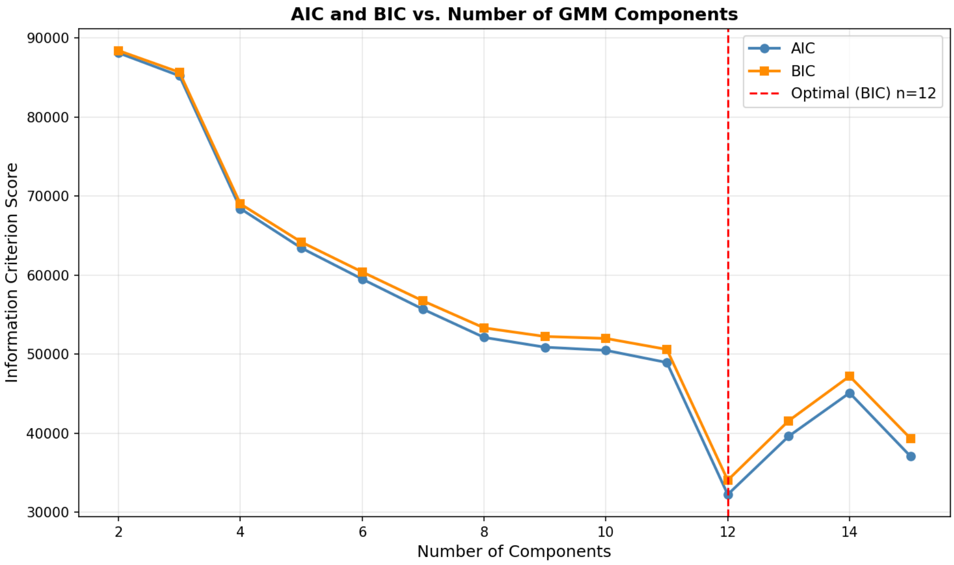

Figure 10: AIC/BIC profile for GMM component selection.

Observed optimum: AIC = 12, BIC = 12 components.

Inference:

- Agreement of AIC and BIC strengthens confidence in $n = 12$.
- Compared with hard clustering methods, GMM better supports overlapping and elliptical seismic distributions.

8.2 GMM world map

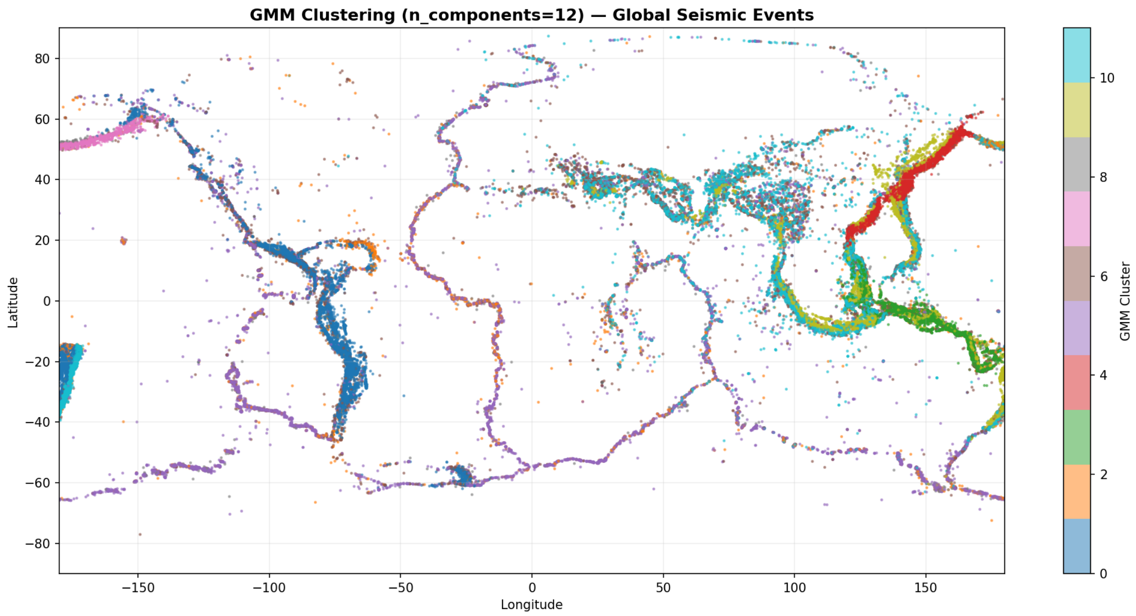

Figure 11: Probabilistic cluster assignments from GMM on global seismic events.

Inference:

- Transition zones are handled more smoothly than in K-Means.
- Tectonically complex regions are modeled with softer boundaries.

9 Cross-Method Visual Comparison

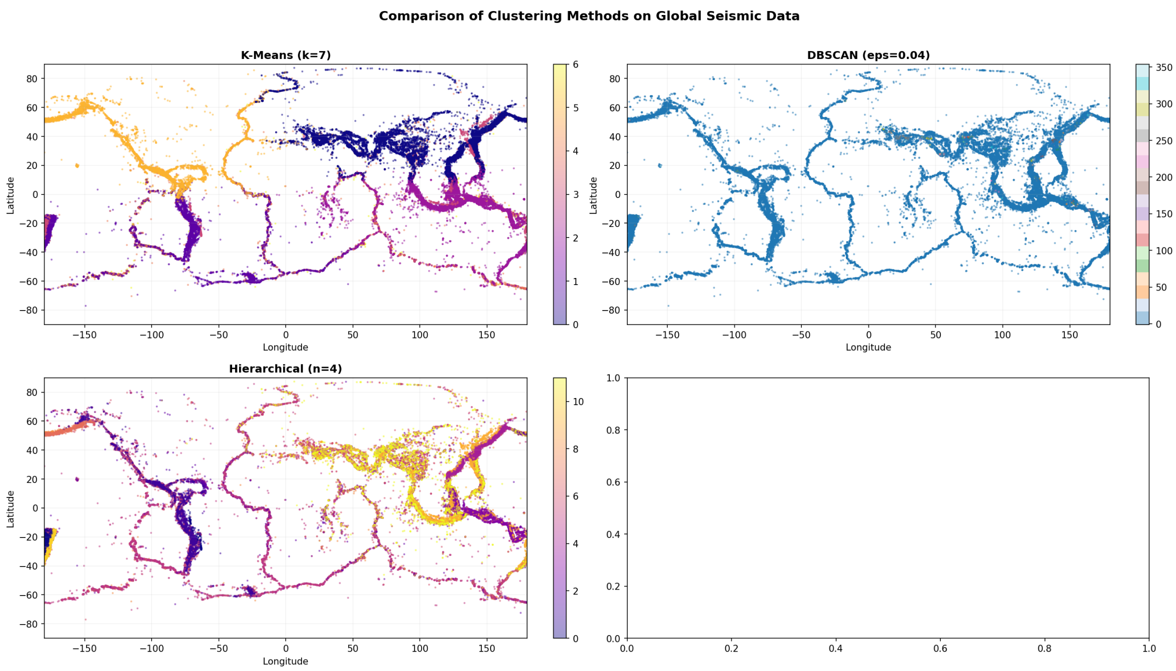

Figure 12: Comparative visualization across clustering methods.

Comparative interpretation:

- **K-Means:** simple, fast, and interpretable, but assumes compact partitions.
- **DBSCAN:** strongest for outlier detection and irregular local density, but highly parameter-sensitive at global scale.
- **Hierarchical:** interpretable multi-level relationships; cluster count can be chosen by both metric and domain view.
- **GMM:** most flexible for overlapping distributions and non-spherical structures.

10 Summary Table

Table 1: Method-wise summary of key outcomes

Method	Main tuning result	Strength	Limitation
K-Means	$k = 7$, silhouette = 0.3432	Fast, clear partitions	Hard boundaries, sp bias
DBSCAN	$\epsilon = 0.04$, 355 clusters	Outlier/density detection	High sensitivity to ϵ
Hierarchical	Best silhouette at $n = 3$	Multi-scale interpretability	Cut-level choice effects
GMM	AIC/BIC optimum at $n = 12$	Overlapping probabilistic	More complex fitting

11 Overall Conclusions

Global seismic activity is concentrated along tectonic plate boundaries and collision zones, especially around the Pacific ring system and major continental interaction belts. Across methods:

- K-Means provides a strong baseline with moderately separated groups.
- DBSCAN highlights sparse versus dense behavior but currently flags very high noise, suggesting further tuning is needed.
- Hierarchical clustering offers interpretable structure and shows a metric-optimal small-cluster solution with optional finer segmentation.
- GMM gives the most realistic probabilistic representation for overlapping seismic regimes.